

ANALYTICAL REPORT

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Laboratory Job ID: 190-28204-1

Client Project/Site: City of Cadillac Biosollids PFAS 2022

For:

City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

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Authorized for release by:
3/23/2022 11:32:22 AM

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Results relate only to the items tested and the sample(s) as received by the laboratory.



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Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-28204-1	DIGESTER #4	Solid	03/08/22 13:30	03/09/22 13:34
190-28204-2	DIGESTER #4 - duplic	Solid	03/08/22 13:30	03/09/22 13:34
190-28204-3	EQUIP. BLANK	Water	03/08/22 13:20	03/09/22 13:34

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Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Job ID: 190-28204-1

Laboratory: Eurofins Michigan

Narrative

Job Narrative 190-28204-1

Comments

No additional comments.

Receipt

The samples were received on 3/9/2022 1:34 PM. Unless otherwise noted below, the samples arrived in good condition, and where required, properly preserved and on ice. The temperature of the cooler at receipt was 5.6° C.

LCMS

Method 537 (modified): Isotope Dilution Analyte (IDA) recovery are above the method recommended limit for the following samples: DIGESTER #4 (190-28204-1), DIGESTER #4 - dublic (190-28204-2), (190-28204-A-2-B MS) and (190-28204-A-2-C MSD). Quantitation by isotope dilution generally precludes any adverse effect on data quality due to elevated IDA recoveries.

Method 537 (modified): The Isotope Dilution Analyte (IDA) recovery associated with the following samples are below the method recommended limit: DIGESTER #4 (190-28204-1), DIGESTER #4 - dublic (190-28204-2), (190-28204-A-2-B MS) and (190-28204-A-2-C MSD). Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the samples.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

General Chemistry

No analytical or quality issues were noted, other than those described in the Definitions/Glossary page.

Organic Prep

Method 3535: Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with preparation batch 320-572353.

320-572353

Method: 3535 PFC-W

Method 3535: Elevated reporting limits are provided for the following sample due to insufficient sample provided for preparation: EQUIP. BLANK (190-28204-3).

320-572353

Method: 3535 PFC-W

Method SHAKE: The following samples were yellow after adjusting to the final volume:
DIGESTER #4 (190-28204-1), DIGESTER #4 - dublic (190-28204-2), (190-28204-A-2 MS) and (190-28204-A-2 MSD)

PFC_IDA

Solid

preparation batch 320-572499

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: DIGESTER #4

Lab Sample ID: 190-28204-1

Date Collected: 03/08/22 13:30

Matrix: Solid

Date Received: 03/09/22 13:34

Percent Solids: 3.0

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
F-53B Major	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
F-53B Minor	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
4:2 FTS	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
6:2 FTS	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
8:2 FTS	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
HFPO-DA (GenX)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	9.4		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorobutanesulfonic acid (PFBS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorobutanoic acid (PFBA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorodecanesulfonic acid (PFDS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorodecanoic acid (PFDA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorododecanoic acid (PFDoA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluoroheptanesulfonic Acid (PFHpS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluoroheptanoic acid (PFHpA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorohexanesulfonic acid (PFHxS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorohexanoic acid (PFHxA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorononanesulfonic acid (PFNS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorononanoic acid (PFNA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorooctanesulfonamide (FOSA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorooctanesulfonic acid (PFOS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorooctanoic acid (PFOA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluoropentanesulfonic acid (PFPeS)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluoropentanoic acid (PFPeA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorotetradecanoic acid (PFTeA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluorotridecanoic acid (PFTriA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1
Perfluoroundecanoic acid (PFUnA)	<6.3		6.3	ug/Kg	☆	03/13/22 20:49	03/15/22 00:39	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	110		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C3 HFPO-DA	91		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C4 PFBA	17	*5-	25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C3 PFBS	103		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C2 PFDA	116		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C2 PFDoA	87		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C4 PFHpA	104		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C2 PFHxA	95		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C5 PFNA	107		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C4 PFOA	108		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C4 PFOS	109		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C5 PFPeA	83		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C2 PFTeA	47		25 - 150	03/13/22 20:49	03/15/22 00:39	1
13C2 PFUnA	117		25 - 150	03/13/22 20:49	03/15/22 00:39	1
d5-NEtFOSAA	128		25 - 150	03/13/22 20:49	03/15/22 00:39	1
d3-NMeFOSAA	117		25 - 150	03/13/22 20:49	03/15/22 00:39	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: DIGESTER #4

Lab Sample ID: 190-28204-1

Date Collected: 03/08/22 13:30

Matrix: Solid

Date Received: 03/09/22 13:34

Percent Solids: 3.0

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
M2-4:2 FTS	128		25 - 150	03/13/22 20:49	03/15/22 00:39	1
M2-6:2 FTS	159	*5+	25 - 150	03/13/22 20:49	03/15/22 00:39	1
M2-8:2 FTS	170	*5+	25 - 150	03/13/22 20:49	03/15/22 00:39	1
18O2 PFHxS	100		25 - 150	03/13/22 20:49	03/15/22 00:39	1

General Chemistry

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Percent Moisture	97.0		0.1	%			03/16/22 15:22	1
Percent Solids	3.0		0.1	%			03/16/22 15:22	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: DIGESTER #4 - duplc

Lab Sample ID: 190-28204-2

Date Collected: 03/08/22 13:30

Matrix: Solid

Date Received: 03/09/22 13:34

Percent Solids: 3.2

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
F-53B Major	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
F-53B Minor	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
4:2 FTS	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
6:2 FTS	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
8:2 FTS	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
HFPO-DA (GenX)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	7.1		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorobutanesulfonic acid (PFBS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorobutanoic acid (PFBA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorodecanesulfonic acid (PFDS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorodecanoic acid (PFDA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorododecanoic acid (PFDoA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluoroheptanesulfonic Acid (PFHpS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluoroheptanoic acid (PFHpA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorohexanesulfonic acid (PFHxS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorohexanoic acid (PFHxA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorononanesulfonic acid (PFNS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorononanoic acid (PFNA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorooctanesulfonamide (FOSA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorooctanesulfonic acid (PFOS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorooctanoic acid (PFOA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluoropentanesulfonic acid (PFPeS)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluoropentanoic acid (PFPeA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorotetradecanoic acid (PFTeA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluorotridecanoic acid (PFTriA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1
Perfluoroundecanoic acid (PFUnA)	<5.8		5.8	ug/Kg	☆	03/13/22 20:49	03/15/22 00:49	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	100		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C3 HFPO-DA	86		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C4 PFBA	21	*5-	25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C3 PFBS	93		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C2 PFDA	108		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C2 PFDoA	83		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C4 PFHpA	86		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C2 PFHxA	97		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C5 PFNA	101		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C4 PFOA	101		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C4 PFOS	103		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C5 PFPeA	76		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C2 PFTeA	39		25 - 150	03/13/22 20:49	03/15/22 00:49	1
13C2 PFUnA	111		25 - 150	03/13/22 20:49	03/15/22 00:49	1
d5-NEtFOSAA	121		25 - 150	03/13/22 20:49	03/15/22 00:49	1
d3-NMeFOSAA	108		25 - 150	03/13/22 20:49	03/15/22 00:49	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: DIGESTER #4 - duplic

Lab Sample ID: 190-28204-2

Date Collected: 03/08/22 13:30

Matrix: Solid

Date Received: 03/09/22 13:34

Percent Solids: 3.2

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
M2-4:2 FTS	119		25 - 150	03/13/22 20:49	03/15/22 00:49	1
M2-6:2 FTS	133		25 - 150	03/13/22 20:49	03/15/22 00:49	1
M2-8:2 FTS	166	*5+	25 - 150	03/13/22 20:49	03/15/22 00:49	1
18O2 PFHxS	91		25 - 150	03/13/22 20:49	03/15/22 00:49	1

General Chemistry

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Percent Moisture	96.8		0.1	%			03/16/22 15:22	1
Percent Solids	3.2		0.1	%			03/16/22 15:22	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: EQUIP. BLANK

Lab Sample ID: 190-28204-3

Date Collected: 03/08/22 13:20

Matrix: Water

Date Received: 03/09/22 13:34

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
F-53B Major	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
F-53B Minor	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
4:2 FTS	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
6:2 FTS	<6.0		6.0	ng/L		03/12/22 07:00	03/16/22 02:07	1
8:2 FTS	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
HFPO-DA (GenX)	<4.8		4.8	ng/L		03/12/22 07:00	03/16/22 02:07	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<6.0		6.0	ng/L		03/12/22 07:00	03/16/22 02:07	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<6.0		6.0	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorobutanesulfonic acid (PFBS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorobutanoic acid (PFBA)	<6.0		6.0	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorodecanesulfonic acid (PFDS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorodecanoic acid (PFDA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorododecanoic acid (PFDoA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluoroheptanoic acid (PFHpA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorohexanesulfonic acid (PFHxS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorohexanoic acid (PFHxA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorononanesulfonic acid (PFNS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorononanoic acid (PFNA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorooctanesulfonamide (FOSA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorooctanesulfonic acid (PFOS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorooctanoic acid (PFOA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluoropentanesulfonic acid (PFPeS)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluoropentanoic acid (PFPeA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorotetradecanoic acid (PFTeA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluorotridecanoic acid (PFTriA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1
Perfluoroundecanoic acid (PFUnA)	<2.4		2.4	ng/L		03/12/22 07:00	03/16/22 02:07	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	89		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C3 HFPO-DA	84		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C4 PFBA	94		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C3 PFBS	99		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C2 PFDA	98		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C2 PFDoA	89		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C4 PFHpA	95		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C2 PFHxA	92		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C5 PFNA	97		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C4 PFOA	93		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C4 PFOS	93		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C5 PFPeA	90		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C2 PFTeDA	87		25 - 150	03/12/22 07:00	03/16/22 02:07	1
13C2 PFUnA	95		25 - 150	03/12/22 07:00	03/16/22 02:07	1
d5-NEtFOSAA	120		25 - 150	03/12/22 07:00	03/16/22 02:07	1
d3-NMeFOSAA	103		25 - 150	03/12/22 07:00	03/16/22 02:07	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: EQUIP. BLANK

Lab Sample ID: 190-28204-3

Date Collected: 03/08/22 13:20

Matrix: Water

Date Received: 03/09/22 13:34

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	126		25 - 150	03/12/22 07:00	03/16/22 02:07	1
M2-6:2 FTS	113		25 - 150	03/12/22 07:00	03/16/22 02:07	1
M2-8:2 FTS	122		25 - 150	03/12/22 07:00	03/16/22 02:07	1
18O2 PFHxS	90		25 - 150	03/12/22 07:00	03/16/22 02:07	1

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 320-572353/1-A

Matrix: Water

Analysis Batch: 573161

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 572353

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
F-53B Major	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
F-53B Minor	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
4:2 FTS	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
6:2 FTS	<5.0		5.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
8:2 FTS	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
HFPO-DA (GenX)	<4.0		4.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<5.0		5.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<5.0		5.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorobutanoic acid (PFBA)	<5.0		5.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorooctanesulfonic acid (PFOS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		03/12/22 07:00	03/16/22 01:07	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	96		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C3 HFPO-DA	89		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C4 PFBA	90		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C3 PFBS	100		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C2 PFDA	103		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C2 PFDoA	91		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C4 PFHpA	94		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C2 PFHxA	96		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C5 PFNA	103		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C4 PFOA	102		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C4 PFOS	101		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C5 PFPeA	96		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C2 PFTeDA	96		25 - 150	03/12/22 07:00	03/16/22 01:07	1
13C2 PFUnA	103		25 - 150	03/12/22 07:00	03/16/22 01:07	1
d5-NEtFOSAA	113		25 - 150	03/12/22 07:00	03/16/22 01:07	1

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-572353/1-A

Matrix: Water

Analysis Batch: 573161

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 572353

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d3-NMeFOSAA	111		25 - 150	03/12/22 07:00	03/16/22 01:07	1
M2-4:2 FTS	105		25 - 150	03/12/22 07:00	03/16/22 01:07	1
M2-6:2 FTS	99		25 - 150	03/12/22 07:00	03/16/22 01:07	1
M2-8:2 FTS	103		25 - 150	03/12/22 07:00	03/16/22 01:07	1
18O2 PFHxS	99		25 - 150	03/12/22 07:00	03/16/22 01:07	1

Lab Sample ID: LCS 320-572353/2-A

Matrix: Water

Analysis Batch: 573161

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 572353

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.7	33.2		ng/L		88	79 - 139
F-53B Major	37.3	32.6		ng/L		87	75 - 135
F-53B Minor	37.7	31.0		ng/L		82	54 - 114
4:2 FTS	37.4	30.2		ng/L		81	79 - 139
6:2 FTS	37.9	41.1		ng/L		108	59 - 175
8:2 FTS	38.3	35.2		ng/L		92	75 - 135
HFPO-DA (GenX)	40.0	37.6		ng/L		94	51 - 173
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	41.0		ng/L		102	76 - 136
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	38.0		ng/L		95	76 - 136
Perfluorobutanesulfonic acid (PFBS)	35.4	31.1		ng/L		88	67 - 127
Perfluorobutanoic acid (PFBA)	40.0	37.1		ng/L		93	76 - 136
Perfluorodecanesulfonic acid (PFDS)	38.6	34.2		ng/L		89	71 - 131
Perfluorodecanoic acid (PFDA)	40.0	36.7		ng/L		92	76 - 136
Perfluorododecanoic acid (PFDoA)	40.0	38.0		ng/L		95	71 - 131
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	32.9		ng/L		86	76 - 136
Perfluoroheptanoic acid (PFHpA)	40.0	34.9		ng/L		87	72 - 132
Perfluorohexanesulfonic acid (PFHxS)	36.4	32.0		ng/L		88	59 - 119
Perfluorohexanoic acid (PFHxA)	40.0	32.5		ng/L		81	73 - 133
Perfluorononanesulfonic acid (PFNS)	38.4	32.8		ng/L		85	75 - 135
Perfluorononanoic acid (PFNA)	40.0	32.3		ng/L		81	75 - 135
Perfluorooctanesulfonamide (FOSA)	40.0	38.0		ng/L		95	73 - 133
Perfluorooctanesulfonic acid (PFOS)	37.1	33.4		ng/L		90	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	35.2		ng/L		88	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.5	33.3		ng/L		89	66 - 126
Perfluoropentanoic acid (PFPeA)	40.0	35.8		ng/L		89	71 - 131
Perfluorotetradecanoic acid (PFTeA)	40.0	34.9		ng/L		87	70 - 130
Perfluorotridecanoic acid (PFTriA)	40.0	40.4		ng/L		101	71 - 131

Eurofins Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-572353/2-A

Matrix: Water

Analysis Batch: 573161

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 572353

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroundecanoic acid (PFUnA)	40.0	39.1		ng/L		98	68 - 128
	LCS	LCS					
Isotope Dilution	%Recovery	Qualifier	Limits				
13C8 FOSA	93		25 - 150				
13C3 HFPO-DA	83		25 - 150				
13C4 PFBA	89		25 - 150				
13C3 PFBS	95		25 - 150				
13C2 PFDA	92		25 - 150				
13C2 PFDoA	88		25 - 150				
13C4 PFHpA	95		25 - 150				
13C2 PFHxA	96		25 - 150				
13C5 PFNA	102		25 - 150				
13C4 PFOA	95		25 - 150				
13C4 PFOS	100		25 - 150				
13C5 PFPeA	87		25 - 150				
13C2 PFTeDA	92		25 - 150				
13C2 PFUnA	95		25 - 150				
d5-NEtFOSAA	95		25 - 150				
d3-NMeFOSAA	105		25 - 150				
M2-4:2 FTS	110		25 - 150				
M2-6:2 FTS	97		25 - 150				
M2-8:2 FTS	99		25 - 150				
18O2 PFHxS	96		25 - 150				

Lab Sample ID: LCSD 320-572353/3-A

Matrix: Water

Analysis Batch: 573161

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 572353

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.7	32.3		ng/L		86	79 - 139	3	30
F-53B Major	37.3	32.7		ng/L		88	75 - 135	1	30
F-53B Minor	37.7	33.3		ng/L		88	54 - 114	7	30
4:2 FTS	37.4	33.7		ng/L		90	79 - 139	11	30
6:2 FTS	37.9	46.8		ng/L		124	59 - 175	13	30
8:2 FTS	38.3	32.5		ng/L		85	75 - 135	8	30
HFPO-DA (GenX)	40.0	39.3		ng/L		98	51 - 173	5	30
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	36.6		ng/L		92	76 - 136	11	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	38.8		ng/L		97	76 - 136	2	30
Perfluorobutanesulfonic acid (PFBS)	35.4	31.9		ng/L		90	67 - 127	2	30
Perfluorobutanoic acid (PFBA)	40.0	35.5		ng/L		89	76 - 136	4	30
Perfluorodecanesulfonic acid (PFDS)	38.6	34.8		ng/L		90	71 - 131	2	30
Perfluorodecanoic acid (PFDA)	40.0	37.3		ng/L		93	76 - 136	2	30
Perfluorododecanoic acid (PFDoA)	40.0	38.6		ng/L		97	71 - 131	2	30

Eurofins Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCSD 320-572353/3-A

Matrix: Water

Analysis Batch: 573161

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 572353

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	36.4		ng/L		96	76 - 136	10	30
Perfluoroheptanoic acid (PFHpA)	40.0	37.7		ng/L		94	72 - 132	8	30
Perfluorohexanesulfonic acid (PFHxS)	36.4	34.2		ng/L		94	59 - 119	6	30
Perfluorohexanoic acid (PFHxA)	40.0	34.7		ng/L		87	73 - 133	7	30
Perfluorononanesulfonic acid (PFNS)	38.4	35.2		ng/L		92	75 - 135	7	30
Perfluorononanoic acid (PFNA)	40.0	34.7		ng/L		87	75 - 135	7	30
Perfluorooctanesulfonamide (FOSA)	40.0	41.3		ng/L		103	73 - 133	8	30
Perfluorooctanesulfonic acid (PFOS)	37.1	35.2		ng/L		95	70 - 130	5	30
Perfluorooctanoic acid (PFOA)	40.0	37.9		ng/L		95	70 - 130	7	30
Perfluoropentanesulfonic acid (PFPeS)	37.5	33.8		ng/L		90	66 - 126	1	30
Perfluoropentanoic acid (PFPeA)	40.0	37.4		ng/L		93	71 - 131	4	30
Perfluorotetradecanoic acid (PFTeA)	40.0	37.1		ng/L		93	70 - 130	6	30
Perfluorotridecanoic acid (PFTriA)	40.0	40.1		ng/L		100	71 - 131	1	30
Perfluoroundecanoic acid (PFUnA)	40.0	37.0		ng/L		93	68 - 128	5	30

Isotope Dilution	LCSD %Recovery	LCSD Qualifier	Limits
13C8 FOSA	95		25 - 150
13C3 HFPO-DA	82		25 - 150
13C4 PFBA	96		25 - 150
13C3 PFBS	98		25 - 150
13C2 PFDA	98		25 - 150
13C2 PFDoA	92		25 - 150
13C4 PFHpA	94		25 - 150
13C2 PFHxA	93		25 - 150
13C5 PFNA	101		25 - 150
13C4 PFOA	92		25 - 150
13C4 PFOS	103		25 - 150
13C5 PFPeA	91		25 - 150
13C2 PFTeDA	98		25 - 150
13C2 PFUnA	97		25 - 150
d5-NEtFOSAA	108		25 - 150
d3-NMeFOSAA	109		25 - 150
M2-4:2 FTS	105		25 - 150
M2-6:2 FTS	102		25 - 150
M2-8:2 FTS	109		25 - 150
18O2 PFHxS	95		25 - 150

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-572499/1-A

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 572499

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
F-53B Major	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
F-53B Minor	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
4:2 FTS	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
6:2 FTS	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
8:2 FTS	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
HFPO-DA (GenX)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorobutanesulfonic acid (PFBS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorobutanoic acid (PFBA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorodecanesulfonic acid (PFDS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorodecanoic acid (PFDA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorododecanoic acid (PFDoA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluoroheptanesulfonic Acid (PFHpS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluoroheptanoic acid (PFHpA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorohexanesulfonic acid (PFHxS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorohexanoic acid (PFHxA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorononanesulfonic acid (PFNS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorononanoic acid (PFNA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorooctanesulfonamide (FOSA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorooctanesulfonic acid (PFOS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorooctanoic acid (PFOA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluoropentanesulfonic acid (PFPeS)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluoropentanoic acid (PFPeA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorotetradecanoic acid (PFTeA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluorotridecanoic acid (PFTriA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1
Perfluoroundecanoic acid (PFUnA)	<0.20		0.20	ug/Kg		03/13/22 20:49	03/14/22 23:48	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	116		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C3 HFPO-DA	96		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C4 PFBA	67		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C3 PFBS	108		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C2 PFDA	112		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C2 PFDoA	113		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C4 PFHpA	104		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C2 PFHxA	99		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C5 PFNA	106		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C4 PFOA	112		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C4 PFOS	112		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C5 PFPeA	98		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C2 PFTeA	115		25 - 150	03/13/22 20:49	03/14/22 23:48	1
13C2 PFUnA	117		25 - 150	03/13/22 20:49	03/14/22 23:48	1
d5-NEtFOSAA	129		25 - 150	03/13/22 20:49	03/14/22 23:48	1

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-572499/1-A

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 572499

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d3-NMeFOSAA	120		25 - 150	03/13/22 20:49	03/14/22 23:48	1
M2-4:2 FTS	111		25 - 150	03/13/22 20:49	03/14/22 23:48	1
M2-6:2 FTS	105		25 - 150	03/13/22 20:49	03/14/22 23:48	1
M2-8:2 FTS	128		25 - 150	03/13/22 20:49	03/14/22 23:48	1
18O2 PFHxS	103		25 - 150	03/13/22 20:49	03/14/22 23:48	1

Lab Sample ID: LCS 320-572499/2-A

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 572499

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	1.88	1.64		ug/Kg		87	79 - 139
F-53B Major	1.86	1.66		ug/Kg		89	74 - 134
F-53B Minor	1.88	1.75		ug/Kg		93	66 - 136
4:2 FTS	1.87	1.87		ug/Kg		100	68 - 143
6:2 FTS	1.90	2.00		ug/Kg		106	73 - 139
8:2 FTS	1.92	1.82		ug/Kg		95	75 - 135
HFPO-DA (GenX)	2.00	2.02		ug/Kg		101	53 - 158
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	2.00	1.87		ug/Kg		93	72 - 132
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	2.00	2.02		ug/Kg		101	72 - 132
Perfluorobutanesulfonic acid (PFBS)	1.77	1.78		ug/Kg		100	69 - 129
Perfluorobutanoic acid (PFBA)	2.00	1.94		ug/Kg		97	76 - 136
Perfluorodecanesulfonic acid (PFDS)	1.93	1.99		ug/Kg		103	71 - 131
Perfluorodecanoic acid (PFDA)	2.00	1.90		ug/Kg		95	72 - 132
Perfluorododecanoic acid (PFDoA)	2.00	1.98		ug/Kg		99	71 - 131
Perfluoroheptanesulfonic Acid (PFHpS)	1.90	1.77		ug/Kg		93	76 - 136
Perfluoroheptanoic acid (PFHpA)	2.00	2.01		ug/Kg		100	71 - 131
Perfluorohexanesulfonic acid (PFHxS)	1.82	1.69		ug/Kg		93	62 - 122
Perfluorohexanoic acid (PFHxA)	2.00	1.90		ug/Kg		95	71 - 131
Perfluorononanesulfonic acid (PFNS)	1.92	1.89		ug/Kg		98	72 - 132
Perfluorononanoic acid (PFNA)	2.00	1.77		ug/Kg		89	73 - 133
Perfluorooctanesulfonamide (FOSA)	2.00	1.99		ug/Kg		99	77 - 137
Perfluorooctanesulfonic acid (PFOS)	1.86	1.84		ug/Kg		99	68 - 141
Perfluorooctanoic acid (PFOA)	2.00	1.87		ug/Kg		94	72 - 132
Perfluoropentanesulfonic acid (PFPeS)	1.88	1.76		ug/Kg		94	66 - 126
Perfluoropentanoic acid (PFPeA)	2.00	1.84		ug/Kg		92	69 - 129
Perfluorotetradecanoic acid (PFTeA)	2.00	1.86		ug/Kg		93	67 - 127
Perfluorotridecanoic acid (PFTriA)	2.00	2.02		ug/Kg		101	71 - 131

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-572499/2-A

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 572499

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroundecanoic acid (PFUnA)	2.00	1.92		ug/Kg		96	66 - 126
	LCS	LCS					
Isotope Dilution	%Recovery	Qualifier	Limits				
13C8 FOSA	109		25 - 150				
13C3 HFPO-DA	92		25 - 150				
13C4 PFBA	60		25 - 150				
13C3 PFBS	100		25 - 150				
13C2 PFDA	114		25 - 150				
13C2 PFDoA	113		25 - 150				
13C4 PFHpA	100		25 - 150				
13C2 PFHxA	94		25 - 150				
13C5 PFNA	108		25 - 150				
13C4 PFOA	104		25 - 150				
13C4 PFOS	109		25 - 150				
13C5 PFPeA	98		25 - 150				
13C2 PFTeDA	106		25 - 150				
13C2 PFUnA	117		25 - 150				
d5-NEtFOSAA	128		25 - 150				
d3-NMeFOSAA	112		25 - 150				
M2-4:2 FTS	108		25 - 150				
M2-6:2 FTS	110		25 - 150				
M2-8:2 FTS	127		25 - 150				
18O2 PFHxS	104		25 - 150				

Lab Sample ID: 190-28204-2 MS

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: DIGESTER #4 - duplic

Prep Type: Total/NA

Prep Batch: 572499

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.8		58.3	51.5		ug/Kg	✱	88	79 - 139
F-53B Major	<5.8		57.7	51.8		ug/Kg	✱	90	74 - 134
F-53B Minor	<5.8		58.3	55.7		ug/Kg	✱	96	66 - 136
4:2 FTS	<5.8		57.8	53.5		ug/Kg	✱	93	68 - 143
6:2 FTS	<5.8		58.6	56.9		ug/Kg	✱	97	73 - 139
8:2 FTS	<5.8		59.3	51.9		ug/Kg	✱	88	75 - 135
HFPO-DA (GenX)	<5.8		61.9	64.0		ug/Kg	✱	103	53 - 158
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<5.8		61.9	59.4		ug/Kg	✱	92	72 - 132
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	7.1		61.9	70.9		ug/Kg	✱	103	72 - 132
Perfluorobutanesulfonic acid (PFBS)	<5.8		54.7	52.3		ug/Kg	✱	96	69 - 129
Perfluorobutanoic acid (PFBA)	<5.8		61.9	61.5		ug/Kg	✱	99	76 - 136
Perfluorodecanesulfonic acid (PFDS)	<5.8		59.6	54.2		ug/Kg	✱	91	71 - 131
Perfluorodecanoic acid (PFDA)	<5.8		61.9	62.7		ug/Kg	✱	101	72 - 132
Perfluorododecanoic acid (PFDoA)	<5.8		61.9	59.7		ug/Kg	✱	96	71 - 131

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: 190-28204-2 MS

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: DIGESTER #4 - duplic

Prep Type: Total/NA

Prep Batch: 572499

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroheptanesulfonic Acid (PFHpS)	<5.8		58.9	54.4		ug/Kg	⊛	92	76 - 136
Perfluoroheptanoic acid (PFHpA)	<5.8		61.9	58.5		ug/Kg	⊛	95	71 - 131
Perfluorohexanesulfonic acid (PFHxS)	<5.8		56.3	55.2		ug/Kg	⊛	98	62 - 122
Perfluorohexanoic acid (PFHxA)	<5.8		61.9	63.3		ug/Kg	⊛	95	71 - 131
Perfluorononanesulfonic acid (PFNS)	<5.8		59.4	54.7		ug/Kg	⊛	92	72 - 132
Perfluorononanoic acid (PFNA)	<5.8		61.9	56.4		ug/Kg	⊛	91	73 - 133
Perfluorooctanesulfonamide (FOSA)	<5.8		61.9	64.4		ug/Kg	⊛	104	77 - 137
Perfluorooctanesulfonic acid (PFOS)	<5.8		57.4	57.4		ug/Kg	⊛	95	68 - 141
Perfluorooctanoic acid (PFOA)	<5.8		61.9	56.7		ug/Kg	⊛	92	72 - 132
Perfluoropentanesulfonic acid (PFPeS)	<5.8		58.0	55.1		ug/Kg	⊛	95	66 - 126
Perfluoropentanoic acid (PFPeA)	<5.8		61.9	55.3		ug/Kg	⊛	89	69 - 129
Perfluorotetradecanoic acid (PFTeA)	<5.8		61.9	60.0		ug/Kg	⊛	97	67 - 127
Perfluorotridecanoic acid (PFTriA)	<5.8		61.9	45.0		ug/Kg	⊛	73	71 - 131
Perfluoroundecanoic acid (PFUnA)	<5.8		61.9	58.9		ug/Kg	⊛	95	66 - 126
Isotope Dilution	MS %Recovery	MS Qualifier	Limits						
13C8 FOSA	114		25 - 150						
13C3 HFPO-DA	89		25 - 150						
13C4 PFBA	18	*5-	25 - 150						
13C3 PFBS	101		25 - 150						
13C2 PFDA	120		25 - 150						
13C2 PFDoA	94		25 - 150						
13C4 PFHpA	102		25 - 150						
13C2 PFHxA	101		25 - 150						
13C5 PFNA	111		25 - 150						
13C4 PFOA	111		25 - 150						
13C4 PFOS	111		25 - 150						
13C5 PFPeA	76		25 - 150						
13C2 PFTeDA	50		25 - 150						
13C2 PFUnA	117		25 - 150						
d5-NEtFOSAA	133		25 - 150						
d3-NMeFOSAA	121		25 - 150						
M2-4:2 FTS	135		25 - 150						
M2-6:2 FTS	159	*5+	25 - 150						
M2-8:2 FTS	197	*5+	25 - 150						
18O2 PFHxS	103		25 - 150						

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: 190-28204-2 MSD

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: DIGESTER #4 - dupli

Prep Type: Total/NA

Prep Batch: 572499

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.8		57.9	47.7		ug/Kg	✱	82	79 - 139	8	30
F-53B Major	<5.8		57.3	52.8		ug/Kg	✱	92	74 - 134	2	30
F-53B Minor	<5.8		57.9	55.2		ug/Kg	✱	95	66 - 136	1	30
4:2 FTS	<5.8		57.4	53.1		ug/Kg	✱	92	68 - 143	1	30
6:2 FTS	<5.8		58.3	67.6		ug/Kg	✱	116	73 - 139	17	30
8:2 FTS	<5.8		58.9	52.7		ug/Kg	✱	89	75 - 135	1	30
HFPO-DA (GenX)	<5.8		61.5	63.7		ug/Kg	✱	104	53 - 158	0	30
N-ethylperfluorooctanesulfonamidoacetic acid (NETFOSAA)	<5.8		61.5	63.0		ug/Kg	✱	99	72 - 132	6	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	7.1		61.5	76.4		ug/Kg	✱	113	72 - 132	7	30
Perfluorobutanesulfonic acid (PFBS)	<5.8		54.4	53.1		ug/Kg	✱	98	69 - 129	2	30
Perfluorobutanoic acid (PFBA)	<5.8		61.5	59.0		ug/Kg	✱	96	76 - 136	4	30
Perfluorodecanesulfonic acid (PFDS)	<5.8		59.3	57.5		ug/Kg	✱	97	71 - 131	6	30
Perfluorodecanoic acid (PFDA)	<5.8		61.5	58.5		ug/Kg	✱	95	72 - 132	7	30
Perfluorododecanoic acid (PFDoA)	<5.8		61.5	61.3		ug/Kg	✱	100	71 - 131	3	30
Perfluoroheptanesulfonic Acid (PFHpS)	<5.8		58.6	55.1		ug/Kg	✱	94	76 - 136	1	30
Perfluoroheptanoic acid (PFHpA)	<5.8		61.5	57.7		ug/Kg	✱	94	71 - 131	1	30
Perfluorohexanesulfonic acid (PFHxS)	<5.8		56.0	57.0		ug/Kg	✱	102	62 - 122	3	30
Perfluorohexanoic acid (PFHxA)	<5.8		61.5	61.0		ug/Kg	✱	92	71 - 131	4	30
Perfluorononanesulfonic acid (PFNS)	<5.8		59.0	55.4		ug/Kg	✱	94	72 - 132	1	30
Perfluorononanoic acid (PFNA)	<5.8		61.5	56.2		ug/Kg	✱	91	73 - 133	0	30
Perfluorooctanesulfonamide (FOSA)	<5.8		61.5	66.2		ug/Kg	✱	108	77 - 137	3	30
Perfluorooctanesulfonic acid (PFOS)	<5.8		57.1	56.4		ug/Kg	✱	94	68 - 141	2	30
Perfluorooctanoic acid (PFOA)	<5.8		61.5	54.6		ug/Kg	✱	89	72 - 132	4	30
Perfluoropentanesulfonic acid (PFPeS)	<5.8		57.7	54.3		ug/Kg	✱	94	66 - 126	1	30
Perfluoropentanoic acid (PFPeA)	<5.8		61.5	62.7		ug/Kg	✱	102	69 - 129	13	30
Perfluorotetradecanoic acid (PFTeA)	<5.8		61.5	60.7		ug/Kg	✱	99	67 - 127	1	30
Perfluorotridecanoic acid (PFTriA)	<5.8		61.5	44.2		ug/Kg	✱	72	71 - 131	2	30
Perfluoroundecanoic acid (PFUnA)	<5.8		61.5	60.5		ug/Kg	✱	98	66 - 126	3	30

Isotope Dilution	MSD %Recovery	MSD Qualifier	Limits
13C8 FOSA	110		25 - 150
13C3 HFPO-DA	93		25 - 150
13C4 PFBA	19	*5-	25 - 150
13C3 PFBS	105		25 - 150
13C2 PFDA	120		25 - 150
13C2 PFDoA	88		25 - 150
13C4 PFHpA	106		25 - 150

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: 190-28204-2 MSD

Matrix: Solid

Analysis Batch: 572885

Client Sample ID: DIGESTER #4 - duplic

Prep Type: Total/NA

Prep Batch: 572499

Isotope Dilution	MSD	MSD	Limits
	%Recovery	Qualifier	
13C2 PFHxA	101		25 - 150
13C5 PFNA	115		25 - 150
13C4 PFOA	115		25 - 150
13C4 PFOS	112		25 - 150
13C5 PFPeA	74		25 - 150
13C2 PFTeDA	43		25 - 150
13C2 PFUnA	118		25 - 150
d5-NEtFOSAA	124		25 - 150
d3-NMeFOSAA	117		25 - 150
M2-4:2 FTS	133		25 - 150
M2-6:2 FTS	156	*5+	25 - 150
M2-8:2 FTS	193	*5+	25 - 150
18O2 PFHxS	104		25 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Solid

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-28204-1	DIGESTER #4	110	91	17 *5-	103	116	87	104	95
190-28204-2	DIGESTER #4 - duplic	100	86	21 *5-	93	108	83	86	97
190-28204-2 MS	DIGESTER #4 - duplic	114	89	18 *5-	101	120	94	102	101
190-28204-2 MSD	DIGESTER #4 - duplic	110	93	19 *5-	105	120	88	106	101
LCS 320-572499/2-A	Lab Control Sample	109	92	60	100	114	113	100	94
MB 320-572499/1-A	Method Blank	116	96	67	108	112	113	104	99

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-28204-1	DIGESTER #4	107	108	109	83	47	117	128	117
190-28204-2	DIGESTER #4 - duplic	101	101	103	76	39	111	121	108
190-28204-2 MS	DIGESTER #4 - duplic	111	111	111	76	50	117	133	121
190-28204-2 MSD	DIGESTER #4 - duplic	115	115	112	74	43	118	124	117
LCS 320-572499/2-A	Lab Control Sample	108	104	109	98	106	117	128	112
MB 320-572499/1-A	Method Blank	106	112	112	98	115	117	129	120

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-28204-1	DIGESTER #4	128	159 *5+	170 *5+	100
190-28204-2	DIGESTER #4 - duplic	119	133	166 *5+	91
190-28204-2 MS	DIGESTER #4 - duplic	135	159 *5+	197 *5+	103
190-28204-2 MSD	DIGESTER #4 - duplic	133	156 *5+	193 *5+	104
LCS 320-572499/2-A	Lab Control Sample	108	110	127	104
MB 320-572499/1-A	Method Blank	111	105	128	103

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Isotope Dilution Summary

Client: City of Cadillac Utilities

Job ID: 190-28204-1

Project/Site: City of Cadillac Biosollids PFAS 2022

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-28204-3	EQUIP. BLANK	89	84	94	99	98	89	95	92
LCS 320-572353/2-A	Lab Control Sample	93	83	89	95	92	88	95	96
LCSD 320-572353/3-A	Lab Control Sample Dup	95	82	96	98	98	92	94	93
MB 320-572353/1-A	Method Blank	96	89	90	100	103	91	94	96

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-28204-3	EQUIP. BLANK	97	93	93	90	87	95	120	103
LCS 320-572353/2-A	Lab Control Sample	102	95	100	87	92	95	95	105
LCSD 320-572353/3-A	Lab Control Sample Dup	101	92	103	91	98	97	108	109
MB 320-572353/1-A	Method Blank	103	102	101	96	96	103	113	111

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-28204-3	EQUIP. BLANK	126	113	122	90
LCS 320-572353/2-A	Lab Control Sample	110	97	99	96
LCSD 320-572353/3-A	Lab Control Sample Dup	105	102	109	95
MB 320-572353/1-A	Method Blank	105	99	103	99

Surrogate Legend

PFOSA = 13C8 FOSA

HFPODA = 13C3 HFPO-DA

PFBA = 13C4 PFBA

C3PFBS = 13C3 PFBS

PFDA = 13C2 PFDA

PFDoA = 13C2 PFDoA

C4PFHA = 13C4 PFHpA

PFHxA = 13C2 PFHxA

PFNA = 13C5 PFNA

PFOA = 13C4 PFOA

PFOS = 13C4 PFOS

PFPeA = 13C5 PFPeA

PFTDA = 13C2 PFTeDA

PFUnA = 13C2 PFUnA

d5NEFOS = d5-NEtFOSAA

d3NMFOS = d3-NMeFOSAA

M242FTS = M2-4:2 FTS

M262FTS = M2-6:2 FTS

M282FTS = M2-8:2 FTS

PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*5-	Isotope dilution analyte is outside acceptance limits, low biased.
*5+	Isotope dilution analyte is outside acceptance limits, high biased.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
α	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CFU	Colony Forming Unit
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MCL	EPA recommended "Maximum Contaminant Level"
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
MPN	Most Probable Number
MQL	Method Quantitation Limit
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
NEG	Negative / Absent
POS	Positive / Present
PQL	Practical Quantitation Limit
PRES	Presumptive
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)
TNTC	Too Numerous To Count

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

LCMS

Prep Batch: 572353

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-28204-3	EQUIP. BLANK	Total/NA	Water	3535	
MB 320-572353/1-A	Method Blank	Total/NA	Water	3535	
LCS 320-572353/2-A	Lab Control Sample	Total/NA	Water	3535	
LCSD 320-572353/3-A	Lab Control Sample Dup	Total/NA	Water	3535	

Prep Batch: 572499

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-28204-1	DIGESTER #4	Total/NA	Solid	SHAKE	
190-28204-2	DIGESTER #4 - duplic	Total/NA	Solid	SHAKE	
MB 320-572499/1-A	Method Blank	Total/NA	Solid	SHAKE	
LCS 320-572499/2-A	Lab Control Sample	Total/NA	Solid	SHAKE	
190-28204-2 MS	DIGESTER #4 - duplic	Total/NA	Solid	SHAKE	
190-28204-2 MSD	DIGESTER #4 - duplic	Total/NA	Solid	SHAKE	

Analysis Batch: 572885

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-28204-1	DIGESTER #4	Total/NA	Solid	537 (modified)	572499
190-28204-2	DIGESTER #4 - duplic	Total/NA	Solid	537 (modified)	572499
MB 320-572499/1-A	Method Blank	Total/NA	Solid	537 (modified)	572499
LCS 320-572499/2-A	Lab Control Sample	Total/NA	Solid	537 (modified)	572499
190-28204-2 MS	DIGESTER #4 - duplic	Total/NA	Solid	537 (modified)	572499
190-28204-2 MSD	DIGESTER #4 - duplic	Total/NA	Solid	537 (modified)	572499

Analysis Batch: 573161

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-28204-3	EQUIP. BLANK	Total/NA	Water	537 (modified)	572353
MB 320-572353/1-A	Method Blank	Total/NA	Water	537 (modified)	572353
LCS 320-572353/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	572353
LCSD 320-572353/3-A	Lab Control Sample Dup	Total/NA	Water	537 (modified)	572353

General Chemistry

Analysis Batch: 573462

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-28204-1	DIGESTER #4	Total/NA	Solid	D 2216	
190-28204-2	DIGESTER #4 - duplic	Total/NA	Solid	D 2216	

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Client Sample ID: DIGESTER #4

Date Collected: 03/08/22 13:30

Date Received: 03/09/22 13:34

Lab Sample ID: 190-28204-1

Matrix: Solid

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Analysis	D 2216		1	573462	03/16/22 15:22	KMW	TAL SAC

Client Sample ID: DIGESTER #4

Date Collected: 03/08/22 13:30

Date Received: 03/09/22 13:34

Lab Sample ID: 190-28204-1

Matrix: Solid

Percent Solids: 3.0

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	SHAKE			572499	03/13/22 20:49	AM	TAL SAC
Total/NA	Analysis	537 (modified)		1	572885	03/15/22 00:39	VPM	TAL SAC

Client Sample ID: DIGESTER #4 - duplc

Date Collected: 03/08/22 13:30

Date Received: 03/09/22 13:34

Lab Sample ID: 190-28204-2

Matrix: Solid

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Analysis	D 2216		1	573462	03/16/22 15:22	KMW	TAL SAC

Client Sample ID: DIGESTER #4 - duplc

Date Collected: 03/08/22 13:30

Date Received: 03/09/22 13:34

Lab Sample ID: 190-28204-2

Matrix: Solid

Percent Solids: 3.2

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	SHAKE			572499	03/13/22 20:49	AM	TAL SAC
Total/NA	Analysis	537 (modified)		1	572885	03/15/22 00:49	VPM	TAL SAC

Client Sample ID: EQUIP. BLANK

Date Collected: 03/08/22 13:20

Date Received: 03/09/22 13:34

Lab Sample ID: 190-28204-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			572353	03/12/22 07:00	EG	TAL SAC
Total/NA	Analysis	537 (modified)		1	573161	03/16/22 02:07	K1S	TAL SAC

Laboratory References:

TAL SAC = Eurofins Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Analyst References:

Lab: TAL SAC

Batch Type: Prep

AM = Andrew Martin

EG = Eric Gomez

Batch Type: Analysis

K1S = Kotechakon Sorndee

KMW = Kelly White

VPM = Veronika Melnik

Eurofins Michigan

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosollids PFAS 2022

Job ID: 190-28204-1

Laboratory: Eurofins Sacramento

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
Alaska (UST)	State	17-020	02-20-24
ANAB	Dept. of Defense ELAP	L2468	01-20-24
ANAB	Dept. of Energy	L2468.01	01-20-24
ANAB	ISO/IEC 17025	L2468	01-20-24
Arizona	State	AZ0708	08-11-22
Arkansas DEQ	State	88-0691	06-17-22
California	State	2897	01-31-23
Colorado	State	CA0004	08-31-22
Florida	NELAP	E87570	06-30-22
Georgia	State	4040	01-30-23
Hawaii	State	<cert No.>	01-29-23
Illinois	NELAP	200060	03-18-22
Louisiana	NELAP	01944	06-30-22
Maine	State	CA00004	04-14-22
Michigan	State	9947	01-29-22 *
Nevada	State	CA00044	08-31-22
New Hampshire	NELAP	2997	04-18-22
New Jersey	NELAP	CA005	06-30-22
New York	NELAP	11666	04-01-22
Ohio	State	41252	01-29-23
Oregon	NELAP	4040	01-29-23
Texas	NELAP	T104704399-19-13	05-31-22
US Fish & Wildlife	US Federal Programs	58448	07-31-22
USDA	US Federal Programs	P330-18-00239	01-23-23
Utah	NELAP	CA000442021-12	03-01-22 *
Virginia	NELAP	460278	03-14-23
Washington	State	C581	05-05-22
West Virginia (DW)	State	9930C	12-31-22
Wisconsin	State	998204680	08-31-22
Wyoming	State Program	8TMS-L	01-28-19 *

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

Eurofins Michigan



Environment Testing
TestAmerica

- ☐ SDS or Known Hazard Information Supplied by Client
☐ Discrepancies
☐ Short Hold
☐ Rush ☐ 24 Hr ☐ 2-Day ☐ 3-Day ☐ 5-Day ☐ Other:
 Receipt Evaluation Performed by: Initials: TEH Date: 3/9/22 Time: 1334

Client ID: City of Cadillac

Work Order #: 190-26204

Cooler / Sample Receipt

After hours receipt: complete gray
areas. Place cooler in walk-in, place
form in Receiving box. Date: _____ Time: _____

Method of Shipment:

Walk-In Client Eurofins TA Field/Courier

Other Client / 3rd Party Courier: _____

Fed Ex Tracking #: _____

UPS Tracking #: Ground

Other: _____

Shipping Container Type:

☒ Cooler ☐ Box

☐ None ☐ Other: _____

Packing Materials:

☒ Plastic Bags ☐ Foam

☐ Bubble Wrap ☐ Paper

☐ Packing Peanuts ☐ None

☐ Other: _____

Custody Seals Intact:

☒ Yes ☐ No

☐ NA (not used or required)

Cooling Materials:

☐ Ice (Solid) ☒ Ice (Melted)

☐ Blue Ice ☐ None

☐ Other: _____

Bacteriological Samples	Temp Corrected (°C)	Frozen?		Rec'd Within 2 Hrs?		Sample Flagged?	
		Yes	No	Yes	No	Yes	No

Received on same day sampled? Yes No

Additional Sheets Required? Yes No

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Acceptable	Cooler ID	Affected Samples
<u>CP313207</u>	<u>5.6</u>	<u>5.6</u>		<u>X</u>	<u>Y</u> <u>N</u>		
					<u>Y</u> <u>N</u>		
					<u>Y</u> <u>N</u>		

Receipt Questions**	Y	N	NA	"No" answers require additional comment
CoC present and ETA receipt signature, date, and time properly documented?	<u>X</u>			
Containers and Labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	<u>X</u>			
Appropriate containers used and adequate volume provided?	<u>X</u>			Preserved bottles checked for pH? Yes No pH strip lot # _____
Number of sample containers match CoC?	<u>X</u>			
Samples received within hold?	<u>X</u>			
Samples submitted for GRO and Volatiles analysis (8260, 624, 524) received without headspace?			<u>X</u>	
Was a Trip Blank received with VOA samples?			<u>X</u>	
Were the samples free of any questionable physical conformities? (i.e.; field duplicates or multiple bottles of the same sample do not significantly vary in appearance - color, solid proportions, etc.)	<u>X</u>			
Were the CoC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	<u>X</u>			
**May not be applicable if samples are not for compliance testing				*Excludes FOG, VOAs, TOC Vials, HEM

Client Contact Record

Contact Via: ☐ Phone ☐ Email ☐ Other: _____ Person Contacted: _____ Date/Time: _____

☐ Discrepancy allowance agreement is on record in the client project file

Discussion / Resolution

Any additional documentation and clarification from the client must be noted in the narrative and/or scanned into the CoC directory.

Reviewed by [Signature] Date: 3/9/22

WI-MI-010_020720

